Variable Resistance Training Methods

Fernando Hernández-Abad

Abstract Variable resistance training devices have become a widespread tool in strength programs. One of the advantages of using this type of training device is the increase or decrease of the external resistance throughout the range of movement. In addition to the good results obtained by the use of elastic bands and chains, which have become widely popular due to their positive adaptations in the expression of strength, versatility, and ease of use, the use of conical pulleys has been added in recent years. In this chapter, we will describe how to approach training with variable resistance training devices, although they are a very ecological resource within the training process and have a wider range of possibilities than the most commonly used. We will take a journey from a traditional vision to offer a more contemporary vision, based on our personal experience, both in high performance and in the rehabilitation and prevention of sports injuries. In addition, there will be a brief introduction and contextualization of the concepts of dynamic rotational stability and vector diversification, proposals to optimize the use of training resources with variable resistance.

Keywords Variable resistance training · Elastic band-resisted training · Dynamic rotational stability training · Vector diversification training · Conic pulley · Force-vector training

1 Introduction

"The development of muscle strength is based on a combination of morphological and neural factors, including the cross-sectional area of the muscle and its architecture, muscle-tendon stiffness, recruitment of motor units, frequency of activation,

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synchronisation of motor units and neuromuscular inhibition" (Suchomel et al. 2018). When designing intervention strategies, there are numerous variables that we use to configure the training load. We have variables such as intensity and volume, which depend on each other and in turn on factors such as speed of contraction, psychological aspects, recovery between sets, order of training contents or frequency of training sessions (Benedict 1999). However, improving an athlete's motor expression, especially in a team sport, with non-reproducible motor situations dependent on opponents, team-mates and various contextual factors, requires other types of variables that help us to equalize the stimuli according to the motor situations that the athlete will face. In this situation, a contextualization of the motor output in absolute terms becomes too reductionist to be efficient. For this reason, and if we understand that "in strength lies the genesis of motricity and therefore in its optimisation lies that of the movements" (Tous 2017), relative values acquire another dimension within the design of intervention strategies and the management of loads in strength training. This forces us to adopt non-conventional variables such as fluctuation of the stimulus; variability of the external load or the biomechanical model of execution; variation in the interserial or repetition range of movement; hierarchical order of activation; and a good number of factors inherent to movement that allow us to adapt the stimuli to athletes' needs. Therefore, both qualitative and quantitative manipulation of any of these factors will affect the others directly or indirectly. The main aim is that such manipulation produces positive adaptations in the body, and by extension in motor performance.

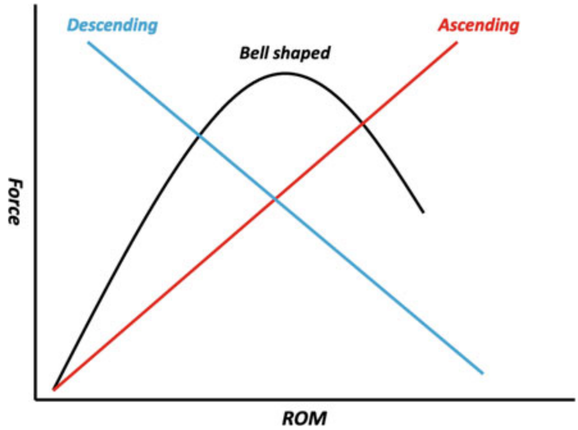
The training devices that can be used to manipulate these non-conventional variables during strength training are those whose resistance is not constant throughout the range of movement. Variable resistance training devices have a resistance that has to be overcome throughout the range of movement (Fleck and Kraemer 2014; McMaster et al. 2009) or per unit of time throughout a repetition. This corresponds to the "criterion of the specificity of the force to the type of movement" (García Manso 1999), making it difficult to think of any sport situation as existing in a stable environment, where the applications of force are constant.

In this chapter we will consider how to approach training with variable resistance devices, which have a wider spectrum of possibilities than the more commonly used devices. We will take a look from a traditional point of view and then offer a more contemporary vision, based on our personal experience, both in high performance and in the rehabilitation and prevention of sports injuries.

2 From Theory

The increase or decrease in resistance throughout the range of movement is one of the advantages of variable resistance training devices. They allow the force–range of movement (ROM) curve to be modified for a proposed exercise. Broadly speaking we can describe 3 main types of force–ROM curve: ascending, descending and bell-shaped (Fleck and Kraemer 2014; Kulig et al. 1984). With the use of

Fig. 1 Curves that commonly represent the expression of force along the range of motion



elastic bands and chains, either exclusively or combined with traditional training methods with free weights and variable resistance tools, we are dealing with an ascending curve (Wallace et al. 2018) (Fig. 1).

The most popular materials for obtaining this type of stimulus are elastic bands, chains or machines with asymmetrical cams (McMaster et al. 2009; Wallace et al. 2018). Although the literature on the effects of training with cam machines is limited, there are numerous studies that demonstrate the positive effects of the use of elastic bands (García-López et al. 2016; Soria-Gila et al. 2015; Kashiani et al. 2020) and chains (Soria-Gila et al. 2015; Kashiani et al. 2020), compared to training using constant resistance (Joy et al. 2016; Rivière et al. 2017; Rhea et al. 2009), and although the use of chains and elastic bands seems to be equally effective (Wallace et al. 2018; Jones 2014), the most popular and widely used are elastic bands. They are easily portable, can be easily attached to various training devices with a high degree of adaptability to the athlete's context, and are affordable. With regard to the parameters of force expression where the best adaptations are obtained by the combined use of free weights and elastic bands, there are studies that refer to significant improvements in the average and peak speed of load displacement and in the average and peak power with respect to constant resistance with free weights (Heelas et al. 2019), as well as in the expression of the force-time slope (Stevenson et al. 2010) and speed in eccentric conditions, which would contribute to improving the efficiency of the stretching-shortening cycle (Wallace et al. 2018). With regard to the training of young athletes, training with elastic bands has been shown to cause similar adaptations in variables such as power, improvements in 1RM and speed of throwing with different positions of the upper limb (Aloui et al. 2019). When athletes are moving loads at high speed, during the concentric phase they spend a good deal of time slowing down the resistance. Variable resistance makes it easier to reduce this deceleration phase (Wallace et al. 2018) in favour of the propulsive phase, encouraging the athlete to apply force for longer during the range of movement. This becomes especially interesting when there is evidence of possible adaptations in reflex-type responses, even improving electromechanical delay (Smith et al. 2019); this is of particular interest, and we consider it to be a field of study that could bring significant advantages in explosive sports, in a type of adaptation that is difficult to improve. However, field experience suggests that the selection of both tools and content must always be subject to their effectiveness in achieving the desired objectives. It has been found that the introduction of this type of variable resistance in certain technical situations can be counterproductive, depending on the objectives (Nijem et al. 2016).

Variable resistance, especially the use of elastic bands, can be a very suitable resource in the early stages of an athlete's return to play after injury. This is especially the case when it comes to avoiding overloading in certain ranges of movement where there is no mechanical advantage and braking the inertia of the movement with an extra load can impose an added risk. Thus, in exercises such as leg presses, squats, or bench presses, the first stages of the concentric phase would not be subjected to as much load as the last ones. This is also true for the eccentric phase, where the first stages would be subjected to more load and the later ones to less, facilitating the eccentric-concentric transition in this type of exercise. The principle of a progressive increase in load is then achieved, not only in the training process or in the return to play process, but also according to the force moment of each phase of the range of movement, which is very important if we want to minimise risks when executing certain exercises. It also seems to be a suitable training tool for improving neuromuscular activation in these processes and when working with particularly sensitive populations such as children or the elderly (Melchiorri and Rainoldi 2011). With respect to this last group, a study has recently been published showing very positive adaptations in a wide variety of functional tests, after three months of training following knee replacement surgery (Liao et al. 2020).

Training using elastic bands or conical pulleys allows movements to be executed bilaterally and unilaterally (Núñez et al. 2018), which is very interesting in terms of rehabilitation processes (Lorenz 2014), optimisation of performance (Núñez et al. 2018; Gonzalo-Skok et al. 2017), and in intervention strategies to compensate for asymmetries between limbs (Núñez et al. 2018).

3 From Practice

During task design it is important to identify risk factors that can generate cognitive biases in the decision-making process. This has led us to think that the structural configuration of training devices conditions our ability to design tasks with a specific objective, regardless of the area of intervention. Training devices with variable resistance, especially elastic and conical-pulley inertial devices, make this process much easier, since they allow us to design stimuli by adapting the force vector and the plane of movement according to the motor expression requirements of the sport (Gonzalo-Skok et al. 2017). From our point of view, these types of

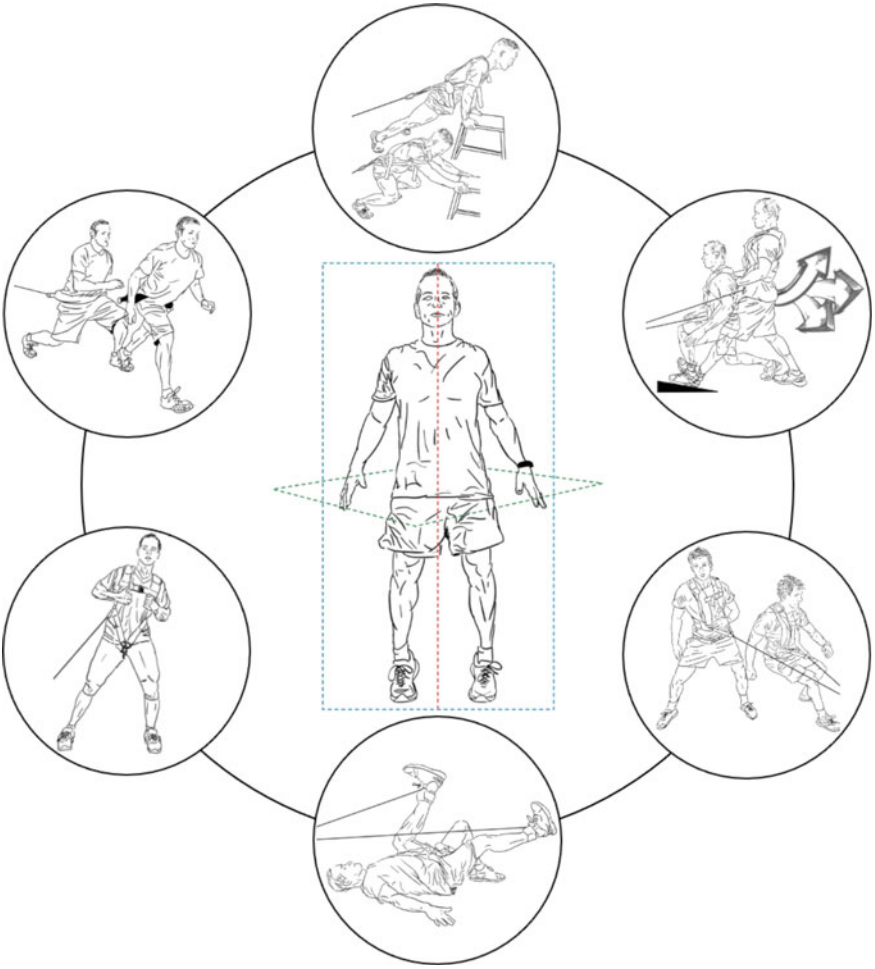


Fig. 2 Some planes and axes of load that can be worked on using variable resistance training

devices enable the design of tasks that are very well adapted to the type of expression of force desired for our athletes, not only with a high degree of dynamic correspondence, but also focusing on any anatomical deficiencies so that they are correctly integrated into the movement (Núñez et al. 2018) (Fig. 2).

Another very important concept that contributes to the optimization of motor output, and which can be efficiently addressed using variable resistance, is dynamic rotational stability. Dynamic rotational stability is the correct response to actions with a rotational component, especially in a monopodal stand, in one or several anatomical areas, in a synergistic way and in the appropriate hierarchical order (Hernandez-Abad 2021). Two joint complexes have generated special interest because of their biomechanical characteristics and the importance of their contribution to mechanical advantage during movement. These are the joint complexes of the hip and the shoulder, also including the myofascial complex that gives stability to the spine, the passive and active structures that shape the plantar arch and their relationship with those previously mentioned. Variable resistance is very beneficial for training in these areas, especially in terms of optimising the training of the sensorimotor system and thus improving neuromuscular responses. This is because of the possibility of working with different planes and non-vertical vectors of force that, during strength training, tend to limit this type of sensorimotor experience because of the biomechanical model used. When using these types of planes and force vectors, it is easier to adopt open biomechanical models (Fig. 3b) that do not need to be executed correctly to avoid triggering the risk of injury, especially when using relatively high resistance accompanied by a high degree of fatigue. This is a major advantage when it comes to adjusting strength training to the structural, neural and metabolic profiles required by athletes.

Variable resistance, in terms of both sports performance and injury prevention, and return to play, offers various advantages that are not always considered:

(1) The possibility of influencing the inclination of the body during the execution of an exercise not only allows net improvement of the propulsive phase of the movement and average speed; it also produces structural, neural and metabolic responses in anatomical areas that optimise performance. In the proposed exercise, engaging the intrinsic and extrinsic muscles of the foot requires greater support with the first metatarsal, providing stability of the plantar arch during execution (Fig. 4a).

(2) Variable resistance can also be used to generate variable perturbation along the ROM, which makes the reflex neuromuscular response of the stabilizing muscles very variable throughout the range (Fig. 4b).

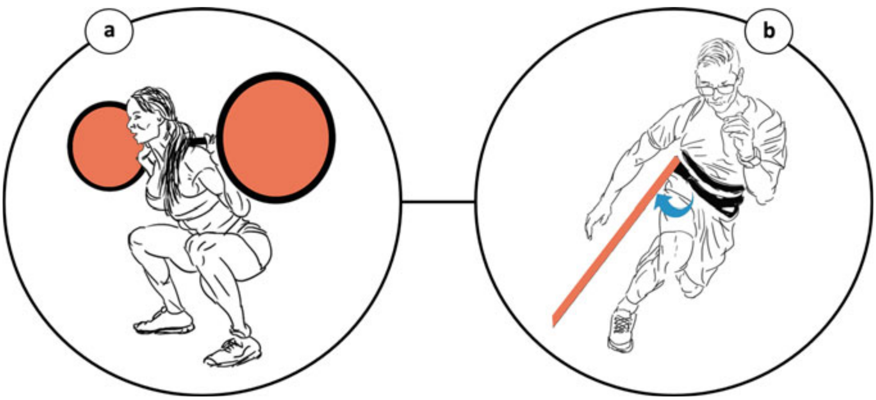


Fig. 3 **a** Closed biomechanical model. When moving high loads, departing from the proposed biomechanical model can lead to a high risk of injury. **b** Open biomechanical model. The proposed biomechanical model is susceptible to exposure to variability without significantly increasing the risk of injury

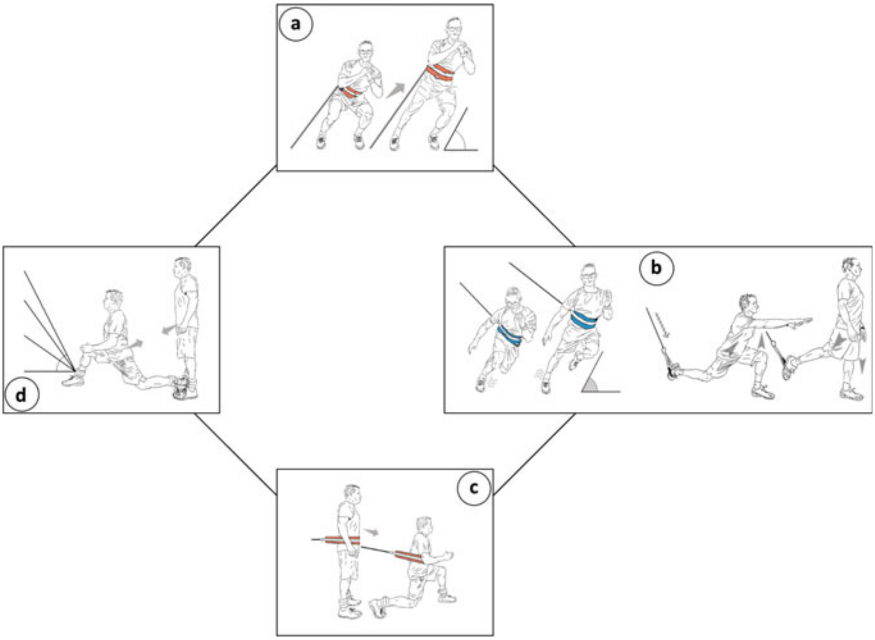


Fig. 4 Some uses of variable resistance in task design, in this example using elastic bands

(3) The management of variable resistance during the execution of an exercise such as that proposed not only generates adaptations during the propulsive phase of the movement, but also dissipates joint stress during landing of the leading leg, which allows reconsideration of the point in the readaptation phase at which this type of movement can be introduced (Fig. 4c).

(4) Using this type of task, not only can we train certain muscles in an eccentric regime with dynamic correspondence, it is also possible to pre-activate appropriate muscles before contact with or support from the ground. In the example proposed, the ischiosural muscles, acting synergistically with the anterior cruciate ligament, provide better knee stability at the start of weight-bearing (Fig. 4d).

Another concept contributed by our working group is that of vector diversification. This involves diversifying a primary force vector into two or more anatomical areas using a small pulley (Fig. 5a). Vector diversification can be considered as a variable when managing the training load with this type of resistance. In our opinion, it represents a step forward in neuromuscular training, since it can facilitate better coactivation of the muscles situated between the anatomical areas to which the primary vector has been diversified. This also promotes the participation of more anatomical areas during the process of pre-activation.

When one of the diversification points is fixed and the other is mobile, a simple hoist is generated, dividing the primary resistance, so that the expression of the

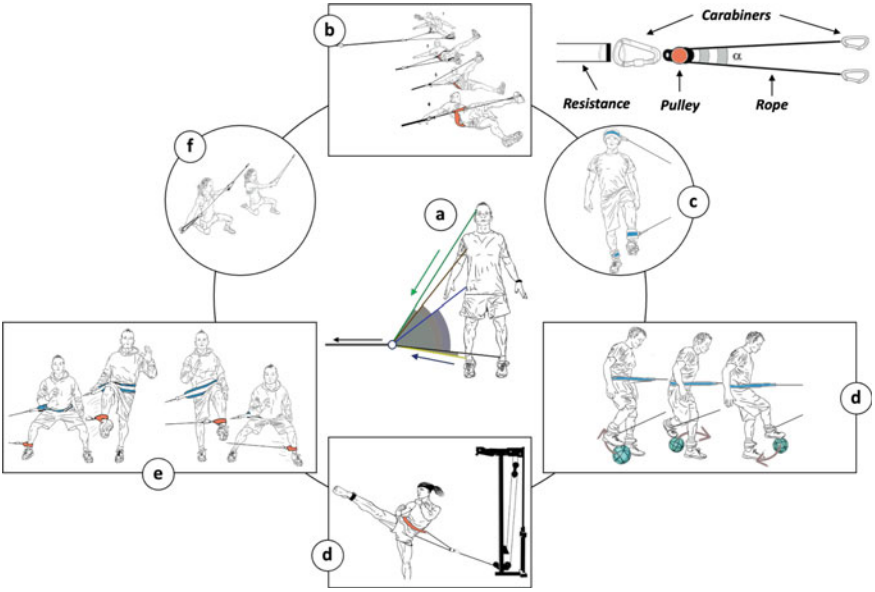


Fig. 5 Examples of vector diversification

force over time is smoothed out, allowing correct technical execution to be maintained at relatively high speeds (Fig. 5d). The distance between the diversified areas determines an angle (α in Fig. 5) that will make the primary resistance divide more or less; the greater the separation between diversified areas, the less the division. Another advantage is that as there is a mobile and a fixed point of attachment, the deformation of the elastic is lower in the same range of movement, and when using weight-guided machines the range of movement will not be limited by the length of the cable of that machine. In this case (Fig. 5d) although we were originally working with a constant resistance machine, the fact that the alpha angle changes along the range of movement converts it into a variable resistance (with limited magnitude), changing the type of stimulus for which the machine is designed. If the angle decreases along the ROM the resistance will be downward (corresponding to the reality of a technical motion such as a kick). If, on the other hand, the angle increases along the ROM the resistance will be upward.

Designing tasks that challenge athletes' neuromuscular responsiveness is relatively easy by diversifying vectors with variable resistance devices. For example, consider a player who executes an open or crossed exit, where one leg applies force in a closed kinetic chain and the other leg does so in an open kinetic chain. By diversifying the vectors, we can add load without affecting the motor action of the leg in the open kinetic chain, while at the same time increasing the rotational moment of force on the leg working in a closed kinetic chain (Fig. 5e).

4 Filling Gaps

During strength training, especially in team sports, the focus has been largely on absolute parameters; net values of a biomechanical pattern with low mechanical and dynamic correspondence to the technical sport movement (Fig. 3a). Variable resistance training equipment is a very useful resource for adapting these net values to the specific characteristics of the sport and its strength requirements. To this end and based on our field experience, it is important to take into account a number of considerations when designing tasks.

4.1 Determining What Type of Variable Resistance Training Device to Use

When deciding which variable resistance device to use, it is important not to solely consider the adaptations in terms of peak speed, power, or length of the propulsive phase. It is also important to take into account other factors, such as the capacity to adapt the training device to the specific planes and axes of load of the sport; the dynamic correspondence; the capacity to generate perturbation during the execution of the exercise; and the potential to provide fluctuation in the external resistance, all without increasing the risk of injury during the execution, independently of the expression of force being worked on (Fig. 6).

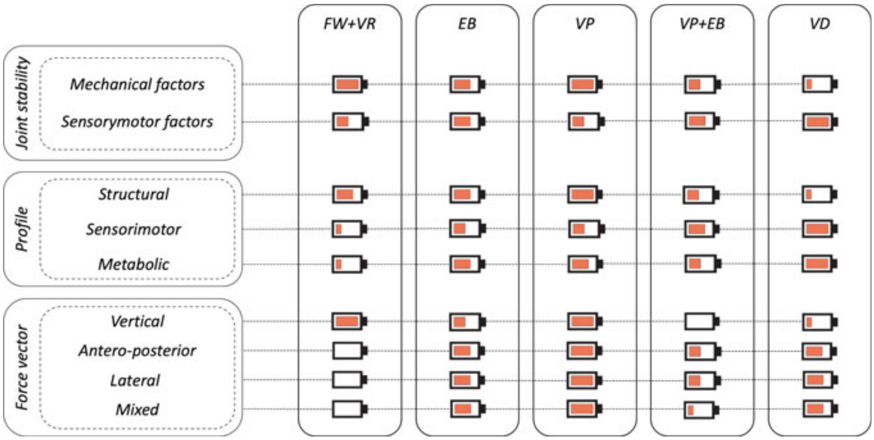


Fig. 6 Variable resistance and adaptability to the target in task design. FW + RV (free weight + variable resistance); EB (elastic bands); VP (versa pulley); VP + EB (versa pulley + variable resistance); VD (vector diversification)

4.2 Considering the Planes and Force Vectors When Designing the Task

With regard to decision making about the planes and vectors of force application during the design of tasks using variable resistance devices, experience tells us that although it is true that when we are searching for certain types of structural adaptations there are non-specific force vectors that can be beneficial, such decision making should be largely influenced by the degree of dynamic correspondence with the sport activity in question. This is important with regard to the integration of anatomical areas in time and form.

4.3 Managing Non-conventional Variables During the Execution of the Task

In the day-to-day life of a physical trainer it is very difficult to control every training variable, and experience tells us that it is also not necessary to have complete control over each variable for a training to be well structured. This chapter has already mentioned some factors that are particularly difficult to measure, but this is not exclusive to strength training sessions; it is also true of technical-tactical training and competitive activity, especially in team sports. From our point of view, with regard to the design of tasks, we should consider how we can intervene in these variables (Fig. 7). These variables may correspond to the competitive activity itself (Fig. 7):

(1) Range of movement within the series itself.
(2) Fluctuation of the external resistance.
(3) Perturbation during repetition (on the training device or on the athlete him/herself).

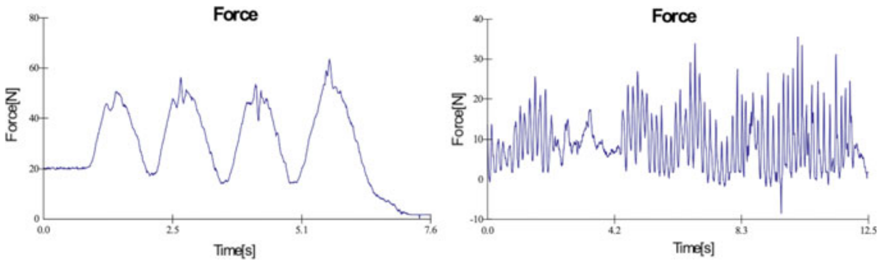


Fig. 7 Example of the dynamometric signal for the same exercise, bilateral standing rowing with elastic + suspension training device, applying a disturbance to the elastic itself during exercise execution (right), and without applying it (left)

(4) Introduction of balls or sports equipment during the execution of the exercise (Moras et al. 2018; Fernández-Valdés et al. 2020).
(5) Dynamic correspondence within the series itself.

4.4 Vector Diversification

When training with variable resistance, conducting generic families of movement in a closed kinetic chain, both elastics and conical pulleys are a very good option. However, when we are working with angular movements of a free extremity, the slope of the force–ROM curve tends to be exponential due to the speed of execution that this type of movement often requires. This, besides carrying a certain risk, doesn't correspond with the characteristics of the resistance to be overcome during the development of competitive motor action, especially for elastics, where at the end of the movement there will be more resistance to overcome. This is where we should consider the hoists used in the diversification of vectors, which can be a very useful tool for smoothing out the force–ROM gradient in this type of exercise, achieving the desired expression of force while keeping athletes within a safe spectrum, especially for those joint complexes that need coactivation of multiple muscles for optimisation of the output.

5 Take-Home Messages and Practical Resources

1. Variable resistance is a very ecological stimulus for athletes, which can not only help to achieve positive adaptations in the propulsive phase of movement, but will also help to dissipate stress in certain anatomical areas, generate selective preactivations, and produce specific structural, neural and metabolic adaptations to the conditional profile of the sport discipline.
2. When working with variable resistances, it is important to consider axes and planes of load in order to be able to orientate the stimulus appropriately, and to take advantage of open biomechanical patterns, which will adapt the mechanical and dynamic correspondence of the proposed task to the sporting action in which performance is to be improved.
3. This correct choice of plane and vector of force and the manipulation of non-conventional variables such as the fluctuation of external resistance; perturbations, variability of the range of movement; and bilateral and unilateral bearings, will promote a better link between the locomotor system, the nervous system and the metabolic system, giving greater specificity to intervention strategies.

4. Dynamic rotational stability is a condition inherent to human motor skills, which is why we believe it should be considered in the strength training process, with vector diversification being a very useful variable during its optimisation, helping us to coactivate more anatomical areas per unit of time.

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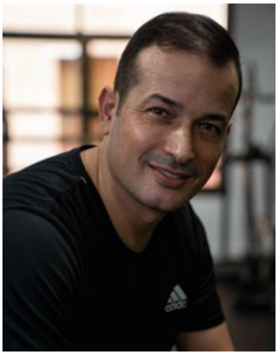

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